1 Choose the Right Plot

I want to...		Plot	Function
See a trend over time		**Line Plot**	`plt.plot() / sns.lineplot()`
Compare categories		**Bar Plot**	`plt.bar() / sns.barplot()`
Count each category		**Count Plot**	`sns.countplot()`
Relation of two numeric features		**Scatter Plot**	`plt.scatter() / sns.scatterplot()`
Distribution of one numeric feature		**Histogram**	`plt.hist() / sns.histplot()`
Spot outliers, compare spread		**Box Plot**	`sns.boxplot()`
Many feature relationships at once		**Pair Plot**	`sns.pairplot()`
Correlation between features		**Heatmap**	`sns.heatmap(df.corr())`

2 EDA Workflow Do the steps in this order

1 Load Data
Read the dataset into a DataFrame

```
pd.read_csv('data.csv')
```

2 Inspect
Shape, columns, data types, summary

```
df.shape   df.info()   df.describe()
```

3 Clean
Missing values, duplicates, wrong types

```
df.isnull().sum()   df.drop_duplicates()
```

4 Single Features
Distribution and outliers of each column

```
sns.histplot()  sns.boxplot()  sns.countplot()
```

5 Relationships
Between features, and with the target

```
sns.scatterplot()  sns.pairplot()  sns.heatmap()
```

6 Preprocess
Fix what the plots revealed

```
impute, handle outliers, encode, scale
```

Ready to build the ML model

3 Common EDA Checks

- [ ] Missing values
- [ ] Duplicate records
- [ ] Outliers (box plot, histogram)
- [ ] Feature distributions
- [ ] Feature relationships
- [ ] Class imbalance (count plot)
- [ ] Highly correlated features
- [ ] Incorrect data types

4 Quick Code Reference

```
sns.lineplot(data=df, x='x', y='y')
sns.barplot(data=df, x='cat', y='val')
sns.scatterplot(data=df, x='x', y='y')
sns.histplot(data=df, x='val', bins=10)
sns.boxplot(data=df, x='cat', y='val')
sns.countplot(data=df, x='cat')
sns.pairplot(df)
sns.heatmap(df.corr(), annot=True)
```

5 Important Notes

- Correlation does not imply causation.
- Plots support EDA but are not a complete analysis.
- Pick the plot by the question and the data type; keep it clean and labeled.